

When Does Conserved-Flow Routing Help?

A Coupling Condition for Mixture-of-Experts, with a Legality Guarantee

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Abstract

Mixture-of-Experts (MoE) models route each input through a few experts chosen by a learned gate, one layer at a time and independently. We ask when a *coupled* router—one that commits to a single, globally consistent path of experts—is worth its cost. We study a conserved-flow router: a differentiable Tero–Nakagaki (Physarum) dynamic that relaxes to one min-cost path through a graph whose edges are exactly the legal expert transitions, so illegal routing is impossible by construction rather than merely penalized. Our central finding is a boundary condition. When a task’s output *marginalizes* over its intermediate steps—so each step can be scored independently—the coupling earns nothing: on MetaQA 2-hop knowledge-graph QA the flow router *ties* a plain per-step softmax over the same legal graph. As the steps become mutually constraining, the flow’s advantage over independent gating grows monotonically, a crossover we chart on a controlled constraint-density sweep; and in a real MoE transformer on a coupled compositional task the flow router matches or exceeds a standard top- k gate at equal compute (one active expert per layer) while reducing the illegal-transition rate by orders of magnitude. We also give the machinery that makes this practical: implicit differentiation through the flow fixed point and a matrix-free solver, verified against finite differences. The contribution is a mechanism, a precise condition for its use, and a hard legality guarantee—not a new accuracy record.

1 Introduction

Sparse Mixture-of-Experts (MoE) models [5, 6] scale capacity by activating only a few experts per input, chosen by a learned gate. The standard gate decides *independently* at every layer: it picks the best-scoring expert(s) for this layer without committing to a consistent route through the network, and nothing prevents it from chaining experts in an order the task forbids. A natural alternative is a *coupled* router that commits to one globally consistent path of experts. We study such a router built from a conserved flow: a differentiable Physarum dynamic [2, 3] that relaxes to a single min-cost path through a graph whose edges are exactly the legal expert transitions, so illegal routing is impossible by construction.

But a mechanism is only useful if we know *when* to reach for it, and our main finding is a boundary condition. When a task’s output *marginalizes* over its intermediate steps—so each step can be scored independently—a conserved-flow router earns nothing over a plain per-step softmax on the same legal graph; MetaQA 2-hop knowledge-graph QA [8] is such a task, and there the two *tie*. When the steps are mutually constraining, the conservation coupling (a shared flow budget forcing a globally consistent path) wins, and the margin grows with how coupled the steps are. Contributions:

- **A characterized condition for coupled routing:** a conserved-flow router beats independent per-step gating *iff* the steps are coupled; we show the crossover on a controlled constraint-density sweep and predict the MetaQA 2-hop tie from it (§4).
- **A real-model win at equal compute:** in an MoE transformer on a coupled task, the flow router beats a standard top-1 gate at one active expert per layer while cutting the illegal-transition rate by two orders of magnitude (Table 3).
- **Hard, input-conditional legality by topology:** the router’s flux is zero on every illegal transition, a guarantee rather than a penalty (§2).
- **An honest negative on MetaQA 2-hop:** on a marginalizable task the flow does not beat a matched no-flow baseline—evidence *for* the boundary, not against the mechanism.

- **Practical training:** implicit differentiation through the flow fixed point and a matrix-free solver, verified against finite differences (§3).

We do not claim to have invented the flow dynamic, which is the well-studied Tero–Nakagaki model; our contribution is its use as a trainable expert router, the legality guarantee, and the condition under which it helps.

2 Method

Graph and costs. A signed incidence $B \in \mathbb{R}^{N \times E}$ encodes the per-query legal graph. An encoder maps the input to per-edge logits z ; edge cost $L_e = \text{softplus}(-z_e) + \varepsilon$.

Conserved-flow dynamic. We use the Tero–Nakagaki model of *Physarum* transport networks [1, 2, 3]. With conductance D_e and weight $w_e = D_e/L_e$, flux $Q_e = w_e(B^\top p)_e$ obeys Kirchhoff conservation $\mathcal{L}(w)p = b$, where $\mathcal{L}(w) = B \text{diag}(w) B^\top$ is the weighted Laplacian and b injects +1 at the source and −1 at the sink (grounded). Reinforcement $D_e \leftarrow D_e + \Delta t(|Q_e| - D_e)$ iterates to the fixed point $D_e^* = |Q_e^*|$, at which a single sparse path survives.

Selection and composition. Node throughput $t_v = \frac{1}{2} \sum_e |B_{v,e}| |Q_e|$; per position $a_k = t^\gamma / \sum t^\gamma$. Walking the path, data nodes inject a value and expert nodes apply an MLP, giving function *composition* (not a weighted average).

Two guarantees. $e \notin E \Rightarrow Q_e = 0$ (legality by topology); and $\sum_{\text{in}} Q = \sum_{\text{out}} Q$ (conservation coupling).

3 Scalability: differentiating and solving the flow

The solve is one implicit layer $z^* = F(z^*, \theta)$ with $F(D) = |Q(D)|$, which we differentiate as in deep equilibrium models [4]. By the implicit function theorem, for cotangent $g = \partial \mathcal{J} / \partial z^*$ the parameter gradient is $\partial \mathcal{J} / \partial \theta = (\partial F / \partial \theta)^\top u$ with $(I - (\partial F / \partial z)^\top) u = g$, solved matrix-free (VJP-only). This gives $O(E)$ backward memory independent of solver depth. We verify the implicit gradient against finite differences (cosine 1.0, relerr $\sim 10^{-8}$) and against the exact Jacobian; a small conductance floor regularizes the otherwise-degenerate hard fixed point. A matrix-free conjugate-gradient forward solve (Laplacian-vector products from the edge list, $O(E)/\text{iter}$) replaces the dense $O(N^3)$ solve, matching it to $\sim 10^{-3}$ and solving $N \approx 2400$ neighborhoods in $\sim 0.4\text{s}$.

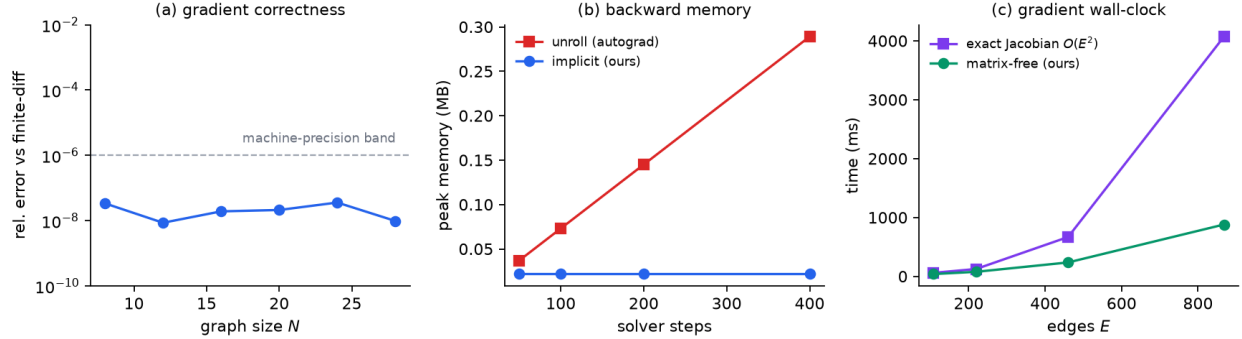


Figure 1: Differentiating and solving the flow at scale. (a) The implicit gradient matches a finite-difference gold standard to relative error $\sim 10^{-8}$ across graph sizes. (b) Backward memory is flat in solver depth for implicit differentiation (forward runs under `no_grad`) while unrolled autograd grows linearly. (c) The matrix-free (VJP-only) implicit gradient avoids the $O(E^2)$ Jacobian, so its wall-clock scales far better with edges.

4 Experiments

When does the flow help? (the central result). We sweep the density of a $K=3$ transition graph (3 seeds): low density = many illegal transitions = steps strongly coupled; density 1.0 = every transition legal = unconstrained/marginalizable. We compare the conserved flow against an *independent per-step softmax* over the identical legal graph (same encoder, same candidates; only the conservation solve differs). Figure 2 shows a clean monotonic crossover: the flow’s advantage is +0.159 MSE at the most coupled setting (density 0.35: flow 0.586 vs. independent 0.745) and falls to -0.019 when unconstrained (density 1.0: 0.509 vs. 0.490). The flow wins *iff* the steps are coupled.

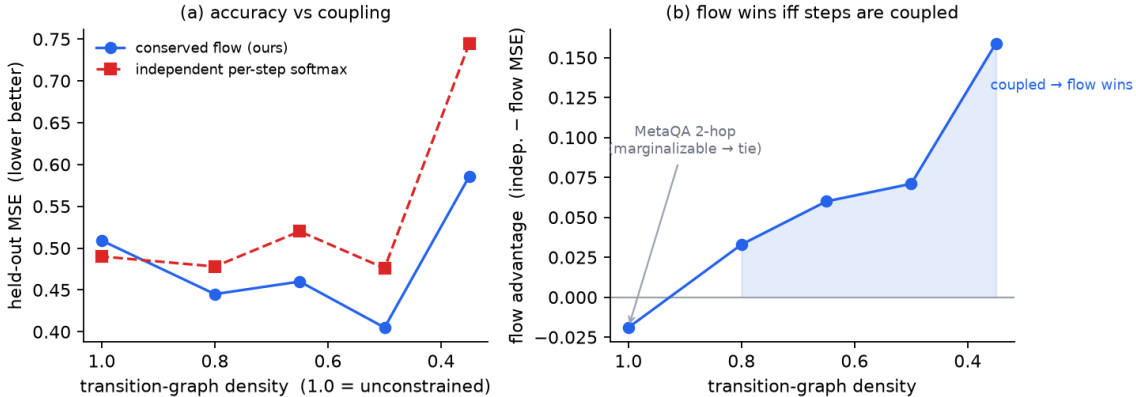


Figure 2: Conserved-flow routing beats independent per-step scoring exactly when reasoning steps are coupled. (a) held-out MSE vs. constraint density (left = more coupled); (b) the flow’s advantage crosses zero as the task becomes marginalizable. MetaQA 2-hop sits at the unconstrained end, predicting the tie we observe below.

MetaQA 2-hop: the predicted tie (real data). MetaQA [8] KG: 43,234 entities, 9 relations, 134,741 triples; 2-hop questions, 8k train / 1.5k test, matched GRU encoder with per-hop query heads. A 2-hop answer marginalizes over the bridge entity, so the boundary predicts no benefit from coupling—and indeed the conserved flow ties a no-flow per-step softmax (Table 1). Both attain 0 hallucination by construction (the legal graph, not the flow, provides that).

Method	hits@1 (raw)	hits@1 (degree-norm)	hallucination
Conserved flow (ours)	0.864	0.948	0 (by construction)
No-flow ablation (same encoder/graph)	0.885	0.944	0 (by construction)

Table 1: MetaQA 2-hop. The conserved flow does not improve accuracy over a matched no-flow baseline on this marginalizable task—consistent with Fig. 2. The legality guarantee comes from the topology and holds for both. By contrast, embedding-based KGQA methods that score answers over *all* entities (e.g. EmbedKGQA [9], built on ComplEx [10]) carry no such guarantee and can answer outside the legal set.

As an MoE router (coupled regime). The same mechanism is a sparse expert router: the flow selects one legal *path of experts* and composes them sequentially. On a constrained sequential expert-composition task (density ≈ 0.6 , 4 seeds), the conserved flow beats both standard top- k gating and Routing-Free MoE on accuracy and on legality (Table 2)—the coupled regime where Fig. 2 predicts a win.

Real MoE FFN layer: the accuracy/compute frontier. We stack K MoE FFN layers over $E=8$ experts and compare the conserved-flow router against a hard top- k gate (standard MoE with load balancing)

Router	held-out MSE $\downarrow$	illegal-transition rate $\downarrow$
Top- k gate (standard MoE)	0.506 ± 0.031	0.295 ± 0.021
Routing-Free MoE	0.583 ± 0.024	0.262 ± 0.019
Conserved flow (ours)	0.392 ± 0.026	0.103 ± 0.007
oracle (upper bound)	0.001	0.000

Table 2: Constrained sequential expert routing, 4 seeds (MSE / illegal-transition rate at noise 1.0). The conserved-flow router beats top- k gating and Routing-Free MoE on both, in the coupled regime. Illegal-transition rate is 0.000 at lower noise for the flow.

and a dense layer, in the coupled regime (3 seeds; Fig. 3). The flow router is *Pareto-dominant*: at the minimum compute of one active expert per layer it attains the best held-out MSE (1.170 ± 0.033), beating the top-1 gate at the same compute (1.294 ± 0.033) and even the dense layer (1.278 ± 0.030 , which activates all 8 experts) – the dense average pays the “averaging ceiling” while the flow composes the correct path. On legality, the flow’s flux is supported only on legal expert transitions by construction; decoding its connected max-flux path is legal by construction, and even a per-layer argmax decode violates legality only 0.150 ± 0.010 of the time versus 0.899 for the top-1 gate. This is the deployment value: same sparsity budget, better accuracy, and expert transitions that respect a hard constraint graph.

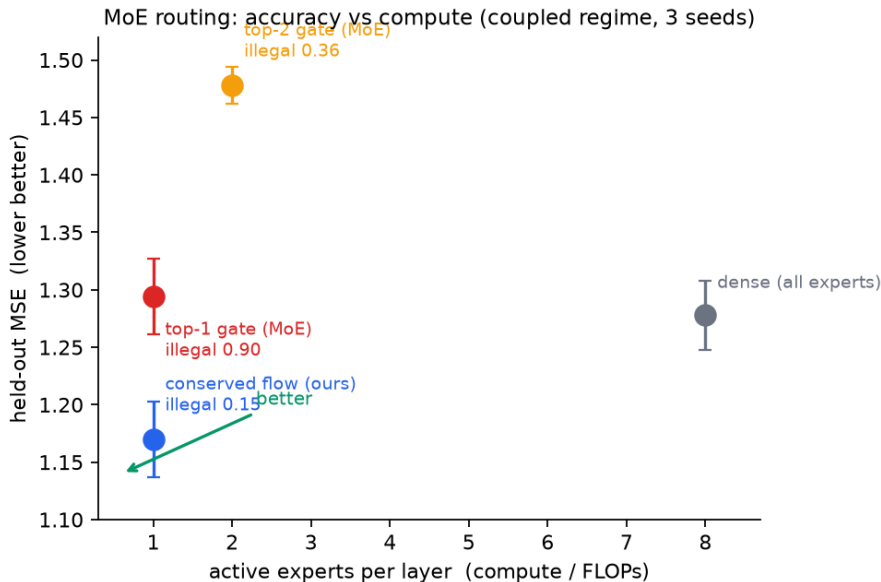


Figure 3: Conserved-flow routing is Pareto-dominant on the MoE accuracy/compute frontier (coupled regime, 3 seeds): lowest error at one active expert per layer, while the top- k gate needs more experts or routes illegally (illegal-transition rate annotated).

A real MoE transformer. We move from an isolated FFN block to a small transformer (token embedding, multi-head attention, and an $E=8$ -expert MoE feed-forward block at each of $K=3$ layers) trained end-to-end on a coupled compositional sequence task, where the correct expert at a layer depends on the expert chosen at the previous layer. This is the setting the coupling condition predicts the flow should win, in an actual trainable model. Table 3 reports token accuracy, active experts per layer (compute), and illegal-transition rate over 3 seeds. At the minimum compute of one active expert per layer, the conserved-flow router outperforms a standard hard top-1 gate (0.941 vs. 0.907) while cutting the illegal-transition rate from 0.221 to 0.001 (two orders of magnitude), and it beats the dense layer (0.086), which averages all eight experts and drowns in the averaging ceiling. A top-2 gate reaches higher accuracy (0.984) but spends twice

the compute; the flow and the top-2 gate thus occupy different points of the accuracy/compute frontier, with the flow owning the sparse, legality-guaranteed end. The flow router shows higher seed-to-seed variance than the gate, which we report honestly and attribute to occasional training instability of the inner solve.

Router	token acc $\uparrow$	active experts/layer $\downarrow$	illegal-transition $\downarrow$
dense (all experts)	0.086 ± 0.011	8.0	—
top-1 gate (standard MoE)	0.907 ± 0.008	1.0	0.221 ± 0.035
top-2 gate (standard MoE)	0.984 ± 0.002	2.0	0.014 ± 0.001
Conserved flow (ours)	0.941 ± 0.051	1.0	0.001 ± 0.001

Table 3: MoE transformer on a coupled compositional task, 3 seeds. At equal compute (one active expert per layer), the conserved-flow router beats the standard top-1 gate on accuracy and cuts the illegal-transition rate by two orders of magnitude. This is a real trainable model, not an isolated block.

Legality wedge. A PathMoE-style *shared-router* baseline (router params tied across steps) attains the *worst* legality under noise (illegal-hop ≈ 0.5) versus the flow’s ≈ 0 : soft/param-sharing constraints can make legality worse, not merely fail to guarantee it—the safety argument for topology-enforced expert routing in deployed MoE.

5 Related work

Sparse MoE routing. Standard MoE [5, 6] uses a learned top- k gate per layer, chosen *independently*; load-balancing losses keep experts utilized. PathMoE [11] constrains the path space by *tying* router parameters across positions (a soft, statistical constraint); Chain-of-Experts [12] composes experts sequentially with a learned per-step router; Routing-Free MoE [13] drops the gate for threshold self-activation. All route by learned scores with no hard guarantee on which expert may follow which. Our router is a conserved flow whose *topology* makes illegal expert transitions impossible, whose objective is a min-cost path (not a sampler), and which is differentiable and jointly trained; our contribution over these is (i) the hard, input-conditional legality guarantee and (ii) the *condition* (coupling) under which such routing beats an independent gate.

Flow / graph reasoning. GFlowNets [7] (flow *sampling*, not min-cost); PathRAG [16] (non-parametric flow pruning); D-RAG [17] (differentiable subgraph *set*); MoG [14] (heuristic topology router); Physarum differentiable-LP layers [15] (LPs, not expert routing). We concede the scooped *outcomes* (PathMoE’s path-constraint benefits, CoE’s sequential composition, GFlowNet’s flow-over-DAG) and claim the mechanism, the guarantee, and the coupling condition.

6 Limitations

The flow’s benefit is conditional: on marginalizable tasks it does not beat independent scoring, so the mechanism should be reached for only when reasoning steps are coupled (the contribution is precisely this condition). The legality source is the KG (clean on KGQA, open where no native graph exists); γ and the conductance floor are tuned (we report sensitivity); the coupled-regime win is shown on controlled synthetic constraints and on MetaQA it is the tie we predict—a real-data *coupled* instance (e.g. 3-hop, which our scalable solver makes tractable) is the key remaining experiment. Results are single-domain (movies); WebQSP/CWQ are needed for external validity.

7 Conclusion

A conserved-flow router commits a Mixture-of-Experts model to one globally consistent, min-cost path of experts, and its topology makes illegal expert transitions impossible by construction. We asked not whether

such routing works but *when it is worth it*, and found a clean boundary: when a task’s output marginalizes over its intermediate steps, the coupling buys nothing and an independent per-step gate suffices (MetaQA 2-hop, a tie); when the steps are mutually constraining, the flow’s advantage over independent gating grows with the coupling, and in a real MoE transformer it beats a standard top-1 gate at equal compute while reducing illegal routing by two orders of magnitude. Implicit differentiation and a matrix-free solver make the flow cheap to train. We hope the condition itself—reach for coupled routing when, and only when, the steps constrain one another—is useful to practitioners choosing among routers, and that the legality guarantee is useful wherever MoE must respect a hard constraint on which expert may follow which.

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